Experiment No-5

Title: Simulation & Demonstration of Pick and Place Robotic System using RoboDK

Aim: To develop, simulate and demonstrate a simple material handling Pick and Place robotic system using RoboDK software

Theory : Material handling robots are widely used in industries for automatic picking and placing of components. A Pick and Place robot uses programmed joint and linear movements to transfer objects from one location to another. RoboDK software provides offline programming and simulation environment for industrial robots, allowing users to create virtual robotic cells before real implementation.

Component Used:

- 1. 6-Axis Industrial Robot
- 2. Robot Controller (Virtual)
- 3. Worktable / Base Frame
- 4. Workpiece (Box/Cylinder)
- 5. Gripper (End Effector)
- 6. Reference Frames
- 7. Targets and Program Instructions

Procedure:

- 1. Open RoboDK and create a New Station.
- 2. Add a 6-axis industrial robot from the online library.
- 3. Insert a worktable and position the robot properly.
- 4. Import the workpiece and place it on the table.
- 5. Add a gripper and attach it to the robot flange (Define TCP).
- 6. Create Pick and Place reference frames.
- 7. Teach required targets: Home, Approach Pick, Pick, Lift, Approach Place, Place, Return Home.
- 8. Create a new program and insert MoveJ and MoveL commands.
- 9. Add Attach and Detach commands to simulate gripping.
- 10. Set speed, acceleration and enable collision detection.
- 11. Run simulation and verify smooth operation.

Program Logic:

- MoveJ Home
- MoveJ Approach Pick

• MoveL Pick

• Attach Object

• MoveL Lift

• MoveJ Approach Place

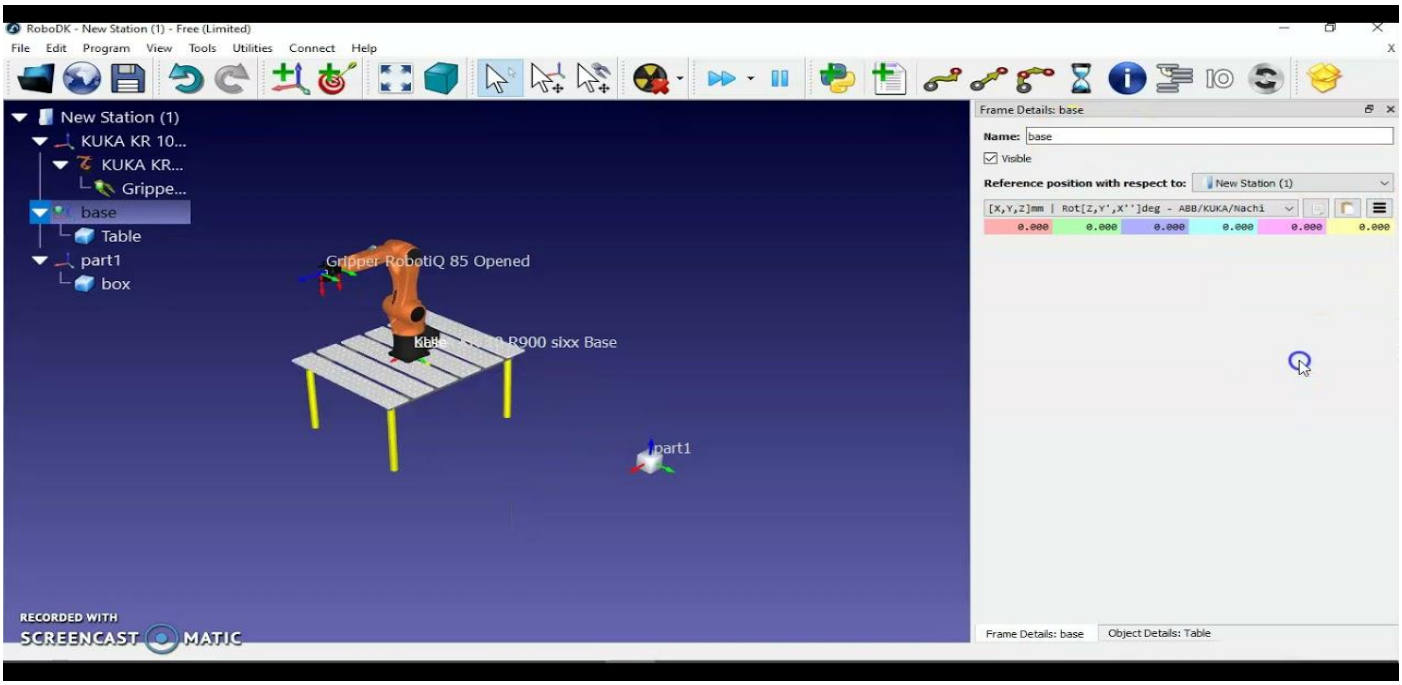
• MoveL Place

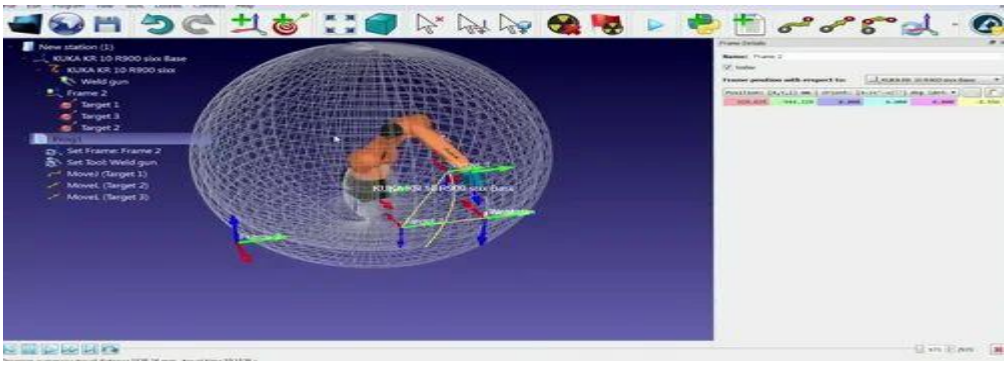
• Detach Object

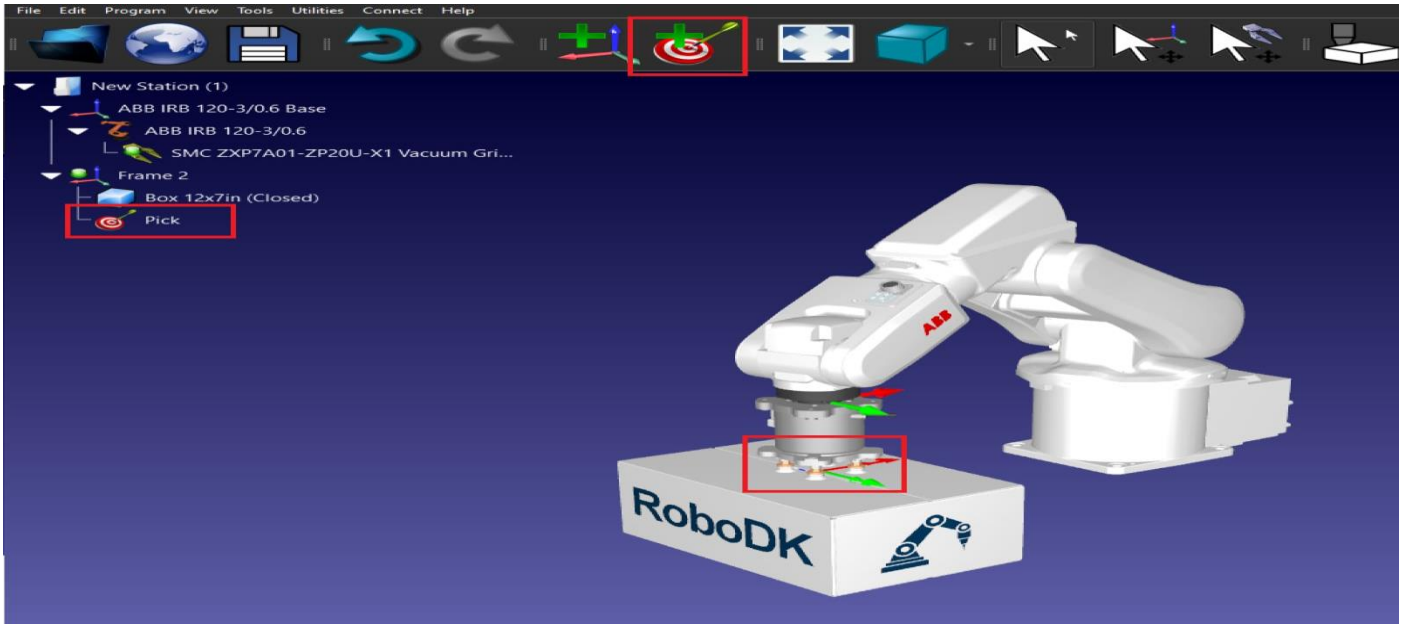
• MoveJ Home

OPERATION FLOW SEQUENCE:

step	operation
1	Robot moves to Home Position
2	Robot approaches Pick Position
3	Robot picks the object
4	Robot lifts the object safely
5	Robot moves to Place Position
6	Robot places the object
7	Robot returns to Home Position







APPLICATIONS:

- Packaging and Sorting Systems
- Assembly Line Automation
- CNC Machine Loading and Unloading
- Pharmaceutical Material Handling
- Food Industry Automation

ADVANTAGES:

- Reduces manual labor
- Improves accuracy and repeatability
- Increases production rate
- Enhances safety in hazardous environments

RESULT: The Pick and Place robotic material handling system was successfully developed and simulated in RoboDK. The robot accurately picked the object from the source location and placed it at the destination location